

On Evaluating the Contribution of Validation for Question Answering

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Abstract—Validation is arising as a crucial component of new architectures aimed at improving Question Answering technologies. Hence, there is a strong need of appropriate measures for evaluating Validation and, even more, its impact on question answering results. However, common Validation measures do not allow a clear study of this impact, and they might even lead researchers to obtain wrong conclusions. We propose a new approach for evaluating Validation technologies, which offers clear and useful information about the impact on QA performance. We compare our proposal with classic evaluation measures, showing the benefits of our scheme.

Index Terms—Question answering, answer validation, evaluation

1 INTRODUCTION

TRADITIONAL Question Answering (QA) systems are based on pipeline architectures, where the typical sequence of steps is: question analysis, document retrieval, passage selection and answer extraction [1]. However, the performance of these architectures is bounded by the dependency among modules and error propagation. This motivated the proposal of new architectures capable of breaking these limitations and producing a qualitative leap in performance [2].

In order to do so, the NLP community tends to promote Answer Validation (AV) systems capable of deciding on the correctness of answers. This idea is behind the architecture of IBM's Watson, which successfully participated at Jeopardy [3]. Watson has a first step focused on over-generating all possible hypotheses with the aim of ensuring correct answers in the searching space. Then, validation focuses on removing incorrect answers and ranking the final candidate ones.

Since validation represents an important component of new QA architectures, it is necessary to use proper and informative evaluation measures for AV modules. Proper evaluation measures must allow comparing different proposals and give useful information about the following criteria:

- 1) Given a pair of AV systems, which contributes most to improve QA results?
- 2) Given an AV system, does it outperform QA results? If not, the AV system must be improved before its inclusion in a real environment.

A suitable evaluation of these criteria would make easier the development of AV modules. However, common validation measures do not allow assessing how validation affects QA results, what may lead researchers to obtain wrong conclusions. This observation led us to set the following research questions:

- 1) What are the best measures, according to the two criteria given above, to study how validation modules contribute to QA results?
- 2) On measuring validation in isolation, do these measures correlate with the expected contribution to QA performance?

In this paper we propose evaluating AV modules using a different method. This method allows studying how validation affects

QA results. We compare our proposal with the classical one, showing the benefits of our approach.

The structure of this paper is: Section 2 reviews previous work related to the evaluation of validation. We show an approach based on QA for evaluating validation in Section 3. Then, we compare this approach with previous proposals in Sections 4 and 5. Finally, some conclusions are given in Section 6.

2 PREVIOUS WORK

Answer validation modules receive a question and an answer from a QA system and decide if the Answer is correct or not [4]. If the module classifies an answer as correct, we say the answer is *validated*. On the other hand, we say that answers considered incorrect are *rejected*. Thus, AV can be seen as a binary classification task where answers are classified as correct or incorrect, with the aim of improving QA results [5].

The validation task gained importance with the first Recognizing Textual Entailment (RTE) Challenge in 2005 [6], where employing RTE systems to validate QA answers was suggested.

Then, the Answer Validation Exercises (AVE) [7] at the Cross Language Evaluation Forum (CLEF) fostered the inclusion of validation in QA architectures. AVE assumed a previous step of over-generation of candidate answers as hypotheses, leaving the hard work to the validation step.

More recently, the Question Answering for Machine Reading Evaluation (QA4MRE) put aside the document retrieval step. The objective was to focus on analysing textual inferences among questions and answers [8]. QA4MRE proposed multiple-choice reading comprehension tests, where each system must detect the correct answer to a question among a set of candidate ones.

2.1 Scenarios for Answer Validation

The expected behaviour of AV modules differs depending on the QA scenario where they are applied. This is because AV modules work on QA systems that may produce different outputs. In this work we focus on AV modules working with factoid QA systems, which has been the main focus of evaluations at the Text REtrieval Conference (TREC) and CLEF. Factoid questions ask about specific facts, as for example the name of a person.

The main types of QA output are:

- *Multiple answers to a question.* We focus here on lists of answers where users search for only one correct answer. There are two kinds of list:
 - *Ranked lists*, where a QA system places answers according to its confidence in answers' correctness. The goal is to obtain a correct answer at the first positions of the list. AV systems in this scenario must place correct answers at the top positions of the list. *Mean Reciprocal Rank (MRR)* [9] is one of the most common QA measures for this scenario.
 - *Non-ranked lists*, where users look for a correct answer in the pool of available ones. AV systems in this scenario must reduce the number of incorrect answers, keeping the correct ones. The objective is to reduce users' effort searching for a correct answer. *Mean Reciprocal Cost (MRC)* [10] has been applied to evaluate QA systems in this scenario.
- *No more than one answer per question.* In this scenario, AV systems must select the final answer from a pool of candidates. *Accuracy* is the most common QA evaluation measure in this scenario. However, it might be possible to left some questions unanswered in order to avoid incorrect answers. In this case, AV systems might have the choice of not selecting any answer. *c@1* is a QA measure proposed for acknowledging

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the reduction of incorrect answers, by leaving some questions unanswered [11].

In this work we focus on systems aiming at reducing the amount of incorrect answers from a pool of candidates given as non-ranked lists.

2.2 Evaluations

AV evaluations use a test collection with questions, answers and human assessments usually obtained from real QA evaluations [7]. AV systems have to classify answers into correct and incorrect ones. Finally, a measure summarizing the quality of the classification with respect to the gold standard collection is applied.

AV systems might find no balance between correct and incorrect candidate answers in real data. This was pointed out by [12], who observed a 78 percent of incorrect answers in the TREC-8 QA output. This must be taken into account when evaluating AV.

Accuracy was one of the first measures proposed given its wide application in binary classification [4], [12]. Nevertheless, *accuracy* presumes that the classes in the target environment are relatively balanced. Thus, a dummy AV module rejecting all the answers may obtain good results in a real environment despite the fact that this result is not informative. This is due to the unbalanced nature of incorrect answers.

Some researchers proposed applying measures focused on evaluating the detection of correct answers, and only them. This is because the main purpose of QA systems, and therefore AV modules, is to return correct answers. This is why the use of *precision* and *recall* were proposed [4]. [7] added the use of the harmonic mean, *F-measure*, of *precision* and *recall* in order to combine them in a measure that trades off both values. *F-measure* has been usually applied as a micro-average.

Results given by *F-measure* depend on the distribution of correct answers in the test collection. This makes harder to compare AV systems across different collections. This is why [13] studied the use of *Receiver Operating Characteristic (ROC)* analysis, which maintains its behaviour when class distribution changes. Despite the positive features of ROC analysis, it was concluded that *F-measure* seemed more informative for evaluating AV modules focus on improving QA results. This is because *F-measure* penalizes in a higher proportion the incorrect validation of answers, which is the worst output that AV modules can produce.

Measures described in this Section evaluate validation as a classification task. However, these measures are not able to give clear information about the impact on QA results. They are more focused on acknowledging the correct classification of answers or/and the detection of correct answers.

3 THE SUGGESTED APPROACH

In this Section we introduce a different approach for evaluating validation over non-ranked lists of answers. This method allows studying how validation affects QA results. A naive approach might be to measure the performance of QA systems including AV modules. However, this is a very expensive and time-consuming activity. Moreover, the actual contribution of a single component is hidden by the complex interaction of all modules.

As it was said above, AV modules classify answers into correct (validated answers) and incorrect (rejected answers). In a real environment, rejected answers are discarded, while the validated ones are returned as the final output of a QA system using the AV module. This output is a list of answers without ranking for each question, as that given by a QA system.

Since this output is similar to the one given by QA systems, we can apply a QA measure over it. Thus, we say that we are evaluating AV systems using a QA approach. This approach has not been applied before.

This method can be applied over the output given by a real QA system, what would require a human assessment of that output (with its associated cost). On the other hand, this approach can be also applied using an AV evaluation collection, already assessed by humans, that reflects the output of real QA systems.

This approach allows researchers to evaluate how validation affects QA results comparing two values:

- the score obtained using the output of the original QA system
- the score obtained after applying the AV module to the QA output, as it has been explained above

Both values are comparable because they represent two different systems over the same set of questions: (1) a traditional QA system, and (2) a QA system using an AV module. A higher score of the combined system would mean that the AV module has contributed to outperform results. However, a lower score of the combined system would mean that the AV module produces a lower performance in the QA system and therefore, the AV module must be improved before using it in a real exploitation environment.

Any QA measure used in scenarios with non-ranked lists can be applied for evaluating AV with this method. We suggest the use of Mean Reciprocal Cost, a measure taken from evaluations at the NTCIR, because it acknowledges the reduction of incorrect answers per question.

The definition of MRC is based on the cost of checking the correctness of answers to a question given a list of n answers, with r correct answers ($c(r,n)$). This cost is defined as follows: firstly, an element of the list is randomly selected and checked. If the selected answer is correct, the cost is 1. If it is not correct, the process is repeated with the other $n-1$ answers, what would have a cost of $c(r, n-1)$. The probability of not having a correct answer in the first attempt is $\frac{n-r}{n}$, which results in the following recurrence formula: $c(r,n) = 1 + c(r, n-1) * \frac{n-r}{n}$. The solution of this formula is $c(r,n) = \frac{n+1}{r+1}$. The value given to each question is the reciprocal of this cost, which is named *Reciprocal Cost*. If $r = 0$, *Reciprocal Cost* is zero. MRC is calculated as the mean of *Reciprocal Cost* over the whole set of questions.

Besides MRC, we have also considered in this work the use of *macro-precision*. We apply the Laplacian correction [14] to avoid divisions by zero when no answer for a given question is validated. Although *macro-precision* is not common in QA, we include it because it acknowledges the correct validation of answers, and gives us more results to study the QA approach. However, one disadvantage produced by the application of the Laplacian correction must be taken into account: An AV module that does not validate any answer would obtained a score of 0.5. This is why we consider the use of MRC more informative.

We have discarded *micro-precision* because it is focused on evaluating the whole set of answers, without paying attention to the performance obtained by each question. Thus, an AV module validating only one answer may obtain a perfect result according to *micro-precision*. This is neither informative nor desirable.

Researchers interested on the QA approach might apply also any other QA measure suitable for the defined scenario.

4 EXPERIMENTAL STUDY

In this Section we answer the second of the research questions proposed in this work. In order to do so, we study whether it is possible to obtain wrong conclusions about how AV modules contribute to QA results, using common validation measures described in Section 2.

We do this study using the QA approach, which has not been applied before for evaluating validation. This method gives a real interpretation of the impact of validation on QA results, since it allows an extrinsic evaluation of AV systems.

In order to do the study, we compare AV systems in an experimental environment created from real data. We evaluate AV systems using two groups of measures:

- *Validation measures* described in Section 2 (we call them AV measures). These measures are: *accuracy*, *precision*, *recall* and its harmonic mean (we use both micro and macro-average), and ROC analysis by means of the *Area Under the Curve* (AUC) measure [15] (we call it ROC in the rest of the paper). These measures evaluate the performance of an AV system as a classifier, using an AV evaluation collection as reference.
- *QA measures* described in Section 3. These measures are *macro-precision* and *MRC*. We perform the evaluation over the output given after applying an AV module to QA responses. The QA system is simulated by the same AV evaluation collection used with the first group of measures. Thus, we are evaluating the application of an AV module to a QA system.

Then, we count situations where the two groups of measures differ on which AV system performs better.

Although *macro-precision* appears in both groups, it is used in two different ways. In the first group of measures, it evaluates the precision of the AV system detecting correct answers. In the second group, it measures the proportion of correct answers, for each question, after applying the AV module.

We prepare one experiment for each one of the two criteria given in Section 1:

- 1) Detecting the best AV system. That is, the system that most contributes to improve QA results.
- 2) Deciding whether an AV system improves or not results of a given QA system.

In next subsections we describe the dataset used and the two experiments.

4.1 Dataset

We take advantage of the AVE 2008 English collection [16]. This dataset was created from the output of real QA systems (those participating at CLEF 2008), and has been used by other researchers [17]. We use this collection as the output of a real QA system whose results must be improved using validation.

The collection contains 1019 answers to 160 questions. There are 79 correct answers and 940 incorrect answers, reflecting the unbalanced nature of QA output.

Furthermore, we automatically create a set of AV runs for the given collection. The purpose is to have a large amount of AV systems available, which permits to perform a high number of comparisons. We create nine different sets of runs. Runs of the same set validate randomly the same percentage of answers, ranking each set from 10 to 90 percent of validated answers, in steps of 10 percent. Each set contains 50 different runs, resulting in a total amount of 450 runs.

4.2 Deciding the Best System

The first experiment is focused on studying wrong interpretations given by the first group of measures (those described in Section 2) when detecting the AV system that most improves QA results. Given two AV systems, we consider a wrong interpretation if: (a) system AVS1 outperforms system AVS2 according to an AV measure, but (b) system AVS2 outperforms system AVS1 according to a QA measure. This means that system AVS2 contributes to obtain a better performance in QA, but the AV measure gives a better ranking to system AVS1. In this case, the AV measure is leading us to a wrong conclusion.

4.2.1 Experiment Description

We compare each pair of AV modules for a given validation percentage. We did not compare all the created runs because of the

TABLE 1
Min-Max Percentages of Wrong Interpretations
When Comparing Pairs of AV Systems

		QA measures	
		macro-P	MRC
	AV measures		
	accuracy	10%-37%	14%-21%
	micro-P	4%-19%	1%-8%
	micro-R	5%-15%	1%-8%
	micro-F	4%-18%	1%-7%
	macro-P	1%-37%	5%-19%
	macro-R	5%-13%	1%-6%
	macro-F	2%-28%	3%-11%
	ROC	4%-19%	1%-8%

high computational cost associated. Given a pair of {AV, QA} measures and the AV systems of a given validation percentage, the sequence of steps is:

- 1) Select each pair of AV systems of that percentage. We have 1225 pairs for each validation percentage.
- 2) Given a pair of AV systems $\{AVS_i, AVS_j\}$, evaluate both systems using the given measures. $eval_{av}(AVS_i)$ represents the score of system AVS_i according to the AV measure, while $eval_{qa}(AVS_i)$ represents the score of system AVS_i according to the QA measure.
- 3) If the two measures differ on which is the best system, that is, if $[eval_{qa}(AVS_i) > eval_{qa}(AVS_j) \text{ AND } eval_{av}(AVS_i) < eval_{av}(AVS_j)]$ OR $[eval_{qa}(AVS_i) < eval_{qa}(AVS_j) \text{ AND } eval_{av}(AVS_i) > eval_{av}(AVS_j)]$, consider it a wrong interpretation of the AV measure. We account only for comparisons with a statistically significant difference using the Wilcoxon test with a confidence of 95 percent.
- 4) Obtain the amount of AV systems pairs of the same set where the two measures differ.

We repeat these steps for each validation percentage given a pair of {AV, QA} evaluation measures.

4.2.2 Results

Table 1 shows the figures for AV systems pairs compared using AV and QA measures. Each cell shows the min-max percentage of wrong interpretations obtained taking into account all the validation percentages.¹ For example, the cell corresponding to row *accuracy* and column *MRC* shows values 14-21 percent. This means that in the set of runs with less pairs of results differing when using *accuracy* and *MRC*, there was a 14 percent of run pairs where the two measures differ. On the other hand, in the set of runs with more pairs of results differing when using *accuracy* and *MRC*, there was a 21 percent of run pairs where the two measures differed. That is, we obtained at least a 14 percent but no more than a 21 percent of wrong conclusions, when using *accuracy* (as a validation measure) instead of *MRC*.

The Table shows how all the validation measures have given a number of wrong interpretations. While *accuracy*, *macro-precision* and *macro-F* were the measures that generated more wrong interpretations, *micro-F* and *ROC* were the ones that produced fewer misunderstandings.

Hence, we conclude that common validation measures can lead researchers to obtain wrong conclusions about the AV system that most contributes to improve QA results. This is why we suggest evaluating validation systems using the QA approach.

1. These min-max values do not come necessarily from sets with the min and max validation percentages. Actually, most of them are obtained from sets with intermediate validation percentages.

TABLE 2
Percentages of Wrong Interpretations When
Comparing AV Systems with a Dummy AV
Baseline

		QA measures	
		macro-P	MRC
	AV measures		
	accuracy	10%	88%
	micro-P	35%	30%
	micro-R	82%	10%
	micro-F	53%	10%
	macro-P	10%	88%
	macro-R	81%	10%
	macro-F	10%	88%
	ROC	35%	28%

4.3 Utility of Validation

The second experiment focuses on studying if researchers can obtain wrong conclusions when deciding whether an AV system improves QA results. That is, if the combination of a QA system with an AV module outperforms results of the single QA system.

4.3.1 Experiment Description

In this experiment we count situations where an AV measure and a QA measure differ about an AV module improving results of a given QA system. We compare each one of the 450 AV runs with a dummy AV system that validates all the answers. This is because the output given by this dummy AV module is the same as that given by the original QA system.

For this purpose, we apply steps 2-4 of Section 4.2.1 for each pair of {AV, QA} evaluation measures.

4.3.2 Results

Table 2 shows the percentages of wrong interpretations obtained for each pair of {AV, QA} measures. Each cell shows the percentage of wrong interpretations obtained using a AV measure instead of a QA measure. For example, the cell corresponding to row *accuracy* and column *MRC* shows value 88 percent. This means that an 88 percent of wrong conclusions about the real contribution of AV systems to QA results has been obtained using *accuracy* instead of *MRC*.

Figures in this table suggest that the use of validation measures can lead researchers to obtain wrong conclusions about the inclusion of validation in QA systems. Therefore, the application of these measures should be avoided for these purposes.

5 TESTING IN REAL ENVIRONMENT

Experiments in Section 4 were performed with systems automatically created from real data. In order to confirm the conclusions obtained, we perform in this Section a similar study using the real AV systems participating at the English AVE 2008 task, where there were eight participants. These systems reflect the behaviour of real AV modules.

5.1 Deciding the Best System

This study is similar to the one described in Section 4.2. For this purpose, we compare each pair of participant AVE systems (56 pairs) using each pair of {AV, QA} evaluation measures. In more detail, we apply steps 2-4 from Section 4.2.1 for each pair of measures.

Table 3 shows the percentages of wrong interpretations achieved. Each cell shows the percentage of wrong comparisons obtained using an AV measure instead of a QA measure. For example, the cell corresponding to row *accuracy* and column *MRC* shows

TABLE 3
Percentages of Wrong Interpretations When
Comparing AV Systems Pairs from AVE 2008

		QA measures	
		macro-P	MRC
	AV measures		
	accuracy	71%	46%
	micro-P	43%	14%
	micro-R	36%	18%
	micro-F	32%	11%
	macro-P	71%	46%
	macro-R	36%	21%
	macro-F	64%	39%
	ROC	25%	7%

value 46 percent. This means that we have obtained a 46 percent of wrong conclusions using *accuracy* instead of *MRC*.

5.2 Utility of Validation

We study in this Section the amount of errors obtained when detecting whether an AV system improves results of a given QA system. For this purpose, we compare the eight participant runs with the baseline AV system that validates all the input answers. We apply steps 2-4 of Section 4.2.1 for each pair of {AV, QA} evaluation measures. The QA system is simulated by the AVE 2008 English test collection, as in Section 4.3.

Table 4 shows results of comparing the AV systems with the proposed baseline. Each cell shows the percentage of wrong conclusions obtained using an AV measure instead of a QA measure. For example, the cell corresponding to row *accuracy* and column *MRC* shows value 38 percent. This means that we have obtained a 38 percent of wrong conclusions about the AV systems that improve results of a QA system.

Despite the fact that the number of comparisons is much lower than in Section 4, the obtained results show the same observations with real systems: *common validation measures might produce wrong conclusions when deciding the contribution of validation to QA results*. For instance, we see how *micro-F* (the main measure in AVE and in other research works) produces some misunderstandings about: (1) which AV module is better, or (2) whether to include or not an AV module in a given QA system.

6 CONCLUSIONS

Question answering validation has arisen as a main component of new architectures aimed at improving results. Since evaluation might contribute to a proper and quicker development of validation technologies, the community is moving towards the proposal of new evaluation tasks.

This paper focuses on QA scenarios where a non-ranked list of answers per question is returned. We have studied how common

TABLE 4
Percentages of Wrong Interpretations When
Comparing AV Systems from AVE 2008 with
a Dummy AV Baseline

		QA measures	
		macro-P	MRC
	AV measures		
	accuracy	25%	38%
	micro-P	0%	25%
	micro-R	63%	63%
	micro-F	0%	13%
	macro-P	25%	38%
	macro-R	63%	63%
	macro-F	25%	38%
	ROC	13%	25%

measures, based on classification, are not able to offer information about how validation affects QA results.

However, if we are more interested on evaluating the impact on QA results, we should apply measures using a QA approach. This method consists of evaluating the output of validation systems as it was the output of QA systems.

We confirm our intuition with some experiments over real data. These experiments show how common validation measures might lead researches to obtain wrong conclusions about: (1) the best performing validation system, and (2) whether a validation system improves or not question answering results.

We have shown that our proposal is able to overcome these drawbacks of common measures. Furthermore, our approach is able to offer clear and useful conclusions about the real contribution of validation.

We do not claim that there exists a measure better than the others. We focus on studying the contribution of validation to QA results, which is important for both validation and QA researchers.

Future work is aimed at testing the proposed method in other scenarios.

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